Project Overview

Interdisciplinary/multidisciplinary description

The overarching goal of this project is to develop a novel AI-based genome annotation tool for genomes of newly sequenced species. The complete DNA blueprint of any living organism is referred to as its *genome*; this information encodes functional molecules called *proteins*. Each protein is encoded in a distinct region called a *gene*. A typical vertebrate has 20,000–30,000 genes; some plants may have as many as 100,000 [37, 40]. However, these genes represent only a very small percentage of the genome: in humans, only about 3% of the 3 billion base pairs of the genome directly encode proteins [17]. Instead, the majority of the genome consists of *repetitive elements* [9, 38] and *introns* (Figure 1), neither of which are used to produce the final protein [25]. Human genes are divided into up to 20 regions (or *exons*) separated by introns that may be thousands of base pairs long. The information necessary to identify the genes in a genome is contained within the sequence of that genome, since it is only that information that is passed from parent to offspring. Hence, the organism is able to use signals in the sequence of its genome to "find" its genes without difficulty. However, researchers have not been able to use those same signals to identify genes because they are very subtle. Thus, the problem of removing the introns and repetitive elements and *annotating* the protein-coding regions of a genome remains an unsolved one in biology and currently requires manual intervention by experts. Here, we propose addressing this genome annotation problem using genome language models (GLMs) – large language models (LLMs) trained on DNA sequences. GLMs have been successfully used in other gene analysis tasks, such as promoter regions prediction, splice site prediction, transcription factor binding site prediction, and the identification of viral variants of interest [3, 14, 41, 10, 42, 5, 2]. Since these models can include hundreds of millions to billions of parameters, their training can be very expensive. To accelerate training and hyperparameter search, we propose using state-of-the-art high-performance computing (HPC) systems, including novel AI accelerators from Cerebras Systems, which have been shown to be effective for both general-purpose LLMs [7] and GLMs [42]. This project brings together three disciplines: *genomics*, *AI*, and *HPC*.

Statement of significance

Obtaining the complete genome of a newly sequenced species enables analyses including structure prediction and phylogenetics. There are already plans to sequence every one of the 70,000 living vertebrate species [35], from sharks to aardvarks. Sequencing these genomes requires: 1) sequencing the DNA on a sequencer, 2) assembling the resulting reads into the genome, and 3) annotating the genome's genes. Huge advances have been made in sequencing technologies and assembly (steps 1&2). Hence, a genome sequence can now be recovered from any organism [40, 21], and sequences from thousands of different species are already available [29, 8]. However, researchers face a serious bottleneck in annotating these genome sequences (step 3). Current annotation approaches require expert intervention and are not scalable to the hundreds of thousands of genomes planned [6]. Nevertheless, the thousands of existing annotated genomes raise the possibility of using LLMs to automatically annotate any *new* genome. Our goal is to collect preliminary data demonstrating the potential of LLMs for annotation in preparation for proposals to NSF or NIH (**Project Description**).

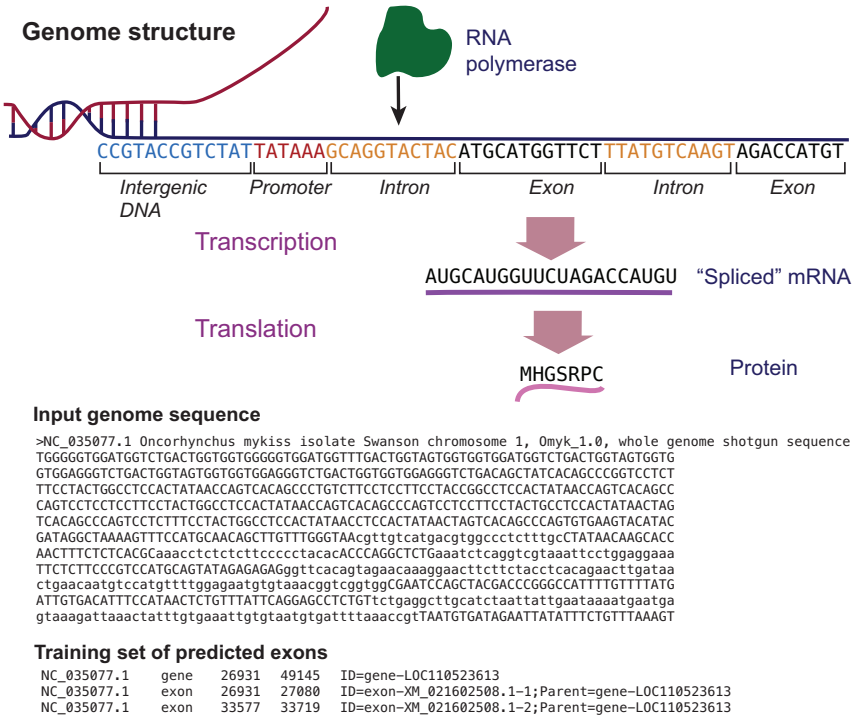

Figure 1: Genomes consist of both non-coding (colored) and coding (black) DNA. The coding exons are transcribed into mRNA molecules, which are then translated into protein sequences. The studied/proposed GLMs will aim to predict the locations of exons in the genome of a given species by leveraging information from other species.

Team description

Michela Becchi, PI, is a faculty member in NCSU's Department of Electrical and Computer Engineering. Her work is broadly in high-performance computing (HPC) and also covers the use of HPC and novel AI accelerators for efficient machine learning at scale [33, 23, 22, 32, 20, 34, 39]. She will advise one of the students working on the project, provide guidance on the effective deployment of the GLMs on GPU clusters and Cerebras systems, and manage the project overall.

Gavin Conant, co-PI, is a faculty member in NCSU's Department of Biological Sciences. His research focuses on the comparison of genomes across many species, and he has been involved in three projects producing new, fully sequenced genomes [31, 36], including leading the project for sequencing the plant *Crambe hispanica* [13]. Co-PI Conant will provide the biological data and background knowledge needed for the project and co-advise the students working on it.

Tianfu Wu, co-PI, is a faculty member in NCSU's Department of Electrical and Computer Engineering. His research is broadly in interpretable and robust machine learning and trustworthy AI, with a focus on interpretable visual modeling, computing, and learning [12, 18]. He has been recently working on fine-tuning LLMs and Vision-Language models (VLMs) for improving performance on downstream tasks with built-in interpretability [28], and on resilient lifelong learning [19, 27]. He will advise a student working on the project for six months and provide guidance on the ML/AI methods used.

Project Description

Problem definition

As a proof of principle for using LLMs in genome annotation, we focus on a specific subproblem: predicting intron–exon boundaries within a chromosome from the genome of a newly sequenced species. Conceptually, we frame this task as a *translation problem*, where the input is an n-character chromosome represented in the four-letter DNA alphabet, and the output is a binary sequence of the same length (0=*exon*, 1=*intron*). The LLM is trained on already annotated genomes from different species.

Hypotheses, objectives and research questions

From a **scientific perspective**, this proposal addresses a high-level research question: Does the naked DNA sequence of a genome contain all the necessary information for an ideal AI tool to accurately predict the set of protein-coding genes from that sequence alone, given sufficient examples as training data? From a **technical perspective**, the project will require addressing questions at the intersection of AI and HPC: (1) How to optimize, adapt and fine-tune existing GLMs to effectively address the problem under consideration? (2) How to address the challenges associated with the length of introns—often 20–30 thousand base pairs (bp), and in rare cases up to 1 million base pairs (e.g., the gene ASIC2 on chromosome 17)? (3) How to effectively deploy the training of large GLMs on traditional distributed HPC systems and emerging AI accelerators, and what are the advantages and disadvantages of each approach?

Technical approach

We will start by fine-tuning existing pre-trained GLMs (designed for other genomics analysis tasks) for the genome annotation problem. The applicability of established GLMs based on the BERT transformer model, such as DNABERT [15] and its extension DNABERT-2 [41], is limited by their input size restrictions (about 500 bp for DNABERT and up to $\sim$10,000 bp for DNABERT-2), which are dictated by their context window—the number of base pairs that can be considered simultaneously. More recent models, such as GENA-LM [10], have extended the context length to handle inputs of up to 36,000 bp using a combination of byte-pair encoding tokenization, sparse attention layers, and recurrent memory transformer architectures. Additionally, the recently announced AlphaGenome [2], trained on human and mouse genomes for a variety of genome analysis tasks, has further expanded the context window to one million bp (although the model has not yet been released). In this work, we will revisit these newer models to study their potential for **annotating genomes from new species using data from existing ones** and we will leverage the power of HPC and emerging AI accelerators for large-scale training and hyperparameter optimization. We will also explore techniques to improve prediction accuracy.

Leveraging repetitive information — Most of the DNA in a typical animal or plant genome is repetitive and therefore uninformative for exon prediction. We propose two methods to alleviate the context window limitation by leveraging this redundancy: (a) the token-to-cluster attention method developed by co-PI Wu [12], and (b) the use of existing LLMs for repeat masking to first mask these repetitive regions, thereby reducing the effective genome size.

Weighting annotation evidence by evolutionary distance — Since our goal is to generate predictions across species, we must take inter-species differences into account. Organisms differ in genome similarity depending on their evolutionary divergence, from closely related species (e.g., humans versus chimpanzees) to more distant ones (e.g., humans versus yeast). We will explore models that overweight evidence for genes based on comparisons to closely related organisms in the training dataset. We have recently developed new tools for performing high-throughput, long-sequence alignments on Cerebras systems, enabling rapid computation of the evolutionary distances required for this approach.

Innovation and novelty

With sequencing and assembly costs dropping, projects aimed at obtaining complete, telomere-to-telomere (T2T) genomes for all living vertebrates are underway [24, 16, 35]. A key bottleneck in these programs is genome annotation, which is currently performed by humans in a non-scalable manner [6]. LLMs have the potential to enable this type of Grand Challenge science. While AI methods have been applied to a range of prediction problems using sequence data, many applications suffer from *limited or poor-quality training data*, making their applicability difficult to assess. At the same time, there is a strong need for annotation tools in the broader community of biologists, and high-quality training data exist from which to build such tools. From a scientific standpoint, these novel AI-based tools will enable comparative biologists to fully harness the ongoing revolution in sequencing technologies. From a technical standpoint, the project will drive innovation in LLM model design, the development of HPC pipelines customized for these applications, and a deeper understanding of the advantages and disadvantages of AI accelerators over traditional HPC systems.

Risks and mitigation strategy

Scientifically, the inability of an LLM to accurately annotate a genome might imply that the contextual signals organisms use to identify genes are more subtle and long-range than currently understood, or that inter-species differences are more significant than expected. An initially unsuccessful approach would therefore still provide valuable feedback for improving AI-based annotation tools. To mitigate risk, we will first focus on predictions among closely related species and then progressively increase species diversity. As more genomes are sequenced, there is strong reason to believe that an AI-based annotator is feasible—especially given that tools such as AlphaFold [4] have already solved the far more biophysically complex problem of protein folding with a moderate amount of training data.

Planning for external funding

We plan to submit proposals to NSF and NIH, building on this work and extending it with novel GLMs, reasoning capabilities, automation, community availability, and broader experimentation. For NSF, we will target the Infrastructure Innovation for Biological Research program (NSF 23-578), which is deadline-free, as is the Plant Genome TRTech-PRG program (NSF 24-547), another suitable venue for this work. For NIH, we will explore funding opportunities through the National Library of Medicine. Additionally, the current DOE focus on the use of foundational models for science, along with their investment in HPC ecosystems and AI testbeds (including Cerebras systems) - `https://www.energy.gov/topics/artificial-intelligence`—suggests upcoming funding opportunities in AI-for-Science and HPC-for-AI. We will leverage our existing collaborations with ANL and PNNL, as well as contacts at ORNL, to get involved in these opportunities.

Data availability

Genome sequences and annotations are available from many sources; however, NIH's GenBank repository is the most comprehensive and currently contains approximately 10,000 annotated eukaryotic genomes from distinct species [30], including 1,500 vertebrate genomes and 700 flowering plant genomes. We will add to these data annotated genomes from Ensembl [8] and Phytozome [11] if they are not present in GenBank set.

Computing resources required

We have allocations on two Cerebras systems: Neocortex at the Pittsburgh Supercomputing Center [26] and the CS-3 system at the AI Testbed at the Argonne Leadership Computing Facility [1]. In addition, we have allocations on multiple NSF-supported ACCESS clusters and cloud platforms (e.g., NCSA Delta CPU and TAMU ACES), which we will use for GPU-based experiments.

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